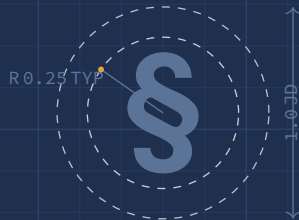


FIELD GUIDE // PART 03 OF 03

LLM METHODS

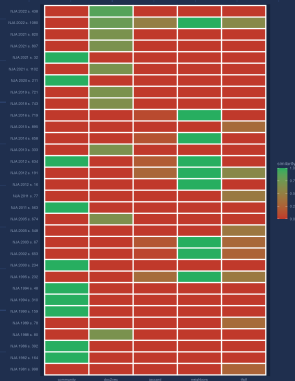
FROM GEOMETRY TO JUDGEMENT



04

Five answers, no agreement

- The five computational readings all answered the same question — and disagreed.
- **32** distinct cases across the five top-10 lists; only **8** appear in more than one list.
- One case (B 5404-22, the A-traktor) is in **4 of 5** lists — the closest thing to consensus.
- Agreement is evidence; it is not truth.



GREEN = IN THE TOP-10, RED = NOT

The sixth reader: an LLM

- All five readings turned law into numbers — and none of the numbers says *why*.
- An LLM can take the **actual texts** as input and return a **verdict with a reason**.
- Same question, new instrument: instead of measuring distance in a space, we *ask* — prompt, context, and a structured answer.
- the workshop's core move: compare the machine's reason against the numbers

verdict, as JSON:

```
{ "similar": true, "reason": "both  
cases concern drunk driving with a  
light vehicle (moped class I)" }
```

the reason is the point --- a
cosine value has no reason

The prompt as a research instrument

- The **system prompt** fixes the task: expert in Swedish law, similarity = same legal question / same rule / analogous facts, answer as JSON only.
- The **user prompt** supplies the two judgments — the anchor's and the candidate's.
- Which text is “the judgment”? A choice: here, the facts from the first instance plus the Supreme Court majority.
- temperature 0, one model, 15 candidates
--- a designed measurement, not a chat

```
# system
"You are an expert in Swedish law ...
similar if they raise the same
legal question, apply the same
rule, or rest on clearly analogous
facts. Respond ONLY with JSON:
{"similar": bool, "reason":
string}"
# user
"ANCHOR JUDGMENT: {{anchor}}
CANDIDATE: {{candidate}} Is it
legally similar? JSON only."
```

What the demo actually does

- Loads the five saved top-10 lists; combines them into one candidate pool; draws the comparison plot.
- For each candidate: build prompt, POST it to an API, parse the JSON verdict, store it.
- **Script practicalities:** the key is in the script (bad practice — keep secrets out of code); loop capped at 15 for time and cost.
- Output: `llm_eval.csv` — case, similar, reason.

```
for (i in 1:15) { prompt <-  
  fill_prompt(cand[i]) resp <-  
  POST(api_url, auth = "Bearer  
    <key>", body = list(model,  
    temperature = 0, messages = ...))  
  verdict <- fromJSON(resp) similar  
    <- verdict$similar reason <-  
    verdict$reason }  
write_csv(llm_results,  
  "llm_eval.csv")
```

What an LLM brings — and what it costs

Gains

- reads *meaning*, not just words
- handles role swap and negation that fool tf-idf
- **justifies** every verdict — auditable
- zero-shot: no training data of its own needed
- Zero-shot LLMs can match trained coders on legal classification (Choi 2024) — but evaluation must be task-specific.
- The reason string is what makes this testable: a human can read it and disagree.

Costs

- **opaque** — we cannot inspect the model
- **non-deterministic** — same prompt, different answers
- **hallucination** — confident wrong reasons
- latency and money; API keys; harder to reproduce

Evaluation: which answer is right?

- No gold standard exists — no court ever published “the similar cases”.
- So we triangulate:
 - **agreement:** cases found by many methods carry more weight;
 - **a reason:** the LLM’s justification can be audited by a lawyer;
 - **a sample:** hand-checking a few verdicts tells you about precision, not just recall.
- Evaluation is a method too — and a legal-theoretical choice.

$\text{precision} = \frac{\text{retrieved \& relevant}}{\text{retrieved}}$

$\text{recall} = \frac{\text{retrieved \& relevant}}{\text{all relevant}}$

in law "relevant" is contested ---
so report and argue, don't just compute

One question, six readings

TEXT **tf-idf / doc2vec**what the opinions say, as vectors

GRAPH **ego / jaccard / community**how the opinions refer, as a graph

LLM **a reasoned verdict**what the opinions mean, in words

- Each reading is a chain of design choices with a price tag: transparency, sparsity, opacity, cost.
- No single reading defines “legally similar” — that is the lawyer’s question, answered with evidence from all six.

Live demo: LLM comparison in R

- Loads the five saved lists; builds compare; replays the tile plot live.
- `get_judgment_text()` selects the facts + majority text for a case; `fill_prompt()` builds the prompt.
- Loops 15 candidates through the API; parses JSON verdicts.
- You get `llm_eval.csv` — and a demonstration of script, auth, prompt, context, output.
- the same script is in your kit

```
$ Rscript 03_llms.R  
[1] combine five top-10 lists 32  
candidates, 8 multi-hit [2] prompt  
& query the API (15 calls, temp 0)  
[3] verdicts -> llm_eval.csv  
{"similar": true, "reason": "..."} ..
```

FIELD GUIDE // END OF SHEET

STOP // QUESTIONS?

ONE CORPUSONE QUESTIONSIX READINGSEACH WITH A COST

